



# Hurricane Emergency Relief System

Software Requirements Specification



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Hurricane Tortilla

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# 1. Introduction

## Description

Hurricanes can have devastating and long-lasting effects on the communities they impact. While immediate concerns such as evacuation, rescue, medical assistance, and emergency shelter are critical, recovery does not end once the storm has passed. Individuals and communities may continue to face challenges such as property damage, debris and cleanup, displaced or missing pets, emotional distress, limited access to necessary supplies, and difficulty finding reliable recovery resources. Some individuals may also require additional assistance before, during, or after a hurricane because of mobility, transportation, accessibility, or other functional needs. With assistance often spread across different organizations and services, finding the right resources can add additional stress to an already difficult recovery process.

## Purpose

The purpose of this project is to create a Hurricane Emergency Response and Relief System that provides a centralized resource for hurricane assistance throughout both the immediate response and recovery periods. The system will help connect victims with emergency response services, shelters, and relief resources while also addressing needs that may continue after immediate safety concerns have been resolved. These secondary needs may include rebuilding and housing resources, community cleanup assistance, pet recovery resources, emotional support services, volunteer opportunities, and other forms of recovery assistance. Rather than replacing existing organizations and services, the system will make these resources easier to locate and access while improving coordination throughout the hurricane response and recovery process.

## Goal

The goal of the project is to make hurricane response and recovery resources easier to find and coordinate for victims, emergency responders, volunteers, and relief organizations. By bringing immediate emergency assistance and longer-term recovery resources together in one system, the project aims to reduce confusion, improve the use of available resources, and support communities throughout all stages of hurricane response and recovery process.

## Scope

This document defines the planned direction and requirements for the Hurricane Emergency Response and Relief System. It identifies the individuals and organizations involved in the system, their needs, and the ways they may interact with hurricane response and recovery services. The document also establishes project constraints and defines the expected capabilities and qualities of the system. Existing emergency response and disaster relief products will also be evaluated to identify successful approaches, existing limitations, and opportunities that can guide the development of this system.

## 2. Stakeholders

- Hurricane Victims and Affected Residents
  - Individuals impacted by a hurricane who may need emergency assistance, shelter, supplies, or recovery resources.
- EMS and First Responders
  - Personnel responsible for responding to urgent requests and assisting individuals facing immediate health or safety risks.
- Shelter Staff and Volunteers
  - Individuals who operate emergency shelters and assist displaced residents with shelter, supplies, and other immediate needs.
- Relief and Humanitarian Organizations
  - Organizations that provide supplies, services, volunteers, and other forms of disaster assistance.
- Community Volunteers
  - Individuals who provide assistance with cleanup, supply distribution, shelter operations, and other relief and recovery efforts.
- Local Government and Emergency Management Agencies
  - Agencies responsible for coordinating hurricane preparedness, emergency response, public information, and community recovery efforts.
- Hospitals and Healthcare Providers
  - Organizations and professionals that provide medical care and other health-related services to affected individuals.
- Animal Shelters and Rescue Organizations
  - Organizations that assist with displaced or lost animals and help reconnect pets with their owners.
- Community Recovery and Support Organizations
  - Organizations that provide services such as cleanup assistance, housing and rebuilding resources, emotional support, and other forms of long-term recovery assistance.

## Personas

### Shelter Volunteer

# MIRANDA COATS

*"I want to spend my time helping people, not wasting it trying to figure out complicated software"*



## QUICK FACTS



Job: Elementary School Teacher



Age: 41



Gender: Female



Address: 1315 Whaley Street, Columbia SC 29205

## ABOUT ME

### Lifestyle

Active in her city and wants to help her community anyway she can. Enjoys spending time with her kids and teaching the youth.

### Personality

Compassionate, patient, dependable, helpful, empathetic, and creative

### Hobbies

Pilates class, hiking with her family, volunteering at the local shelter, and watching movies

## EDUCATION & EXPERIENCE

### ✓ Education

Bachelor's Degree in Elementary Education from University of South Carolina

### ✓ Technology Proficiency

Moderate technological ability with everyday apps and platforms, but not with specialized emergency service software.

### ✓ Technology Platforms

Personal Computer, Iphone, and Ipad

## GOALS & FRUSTRATIONS

### ► GOALS AND NEEDS

#### Goals:

Be able to help people during hurricane emergencies to the best of her ability. Able to be impactful in hurricane relief while still working a job and having a family.

#### Needs:

A system that is easy to use and understand for someone who does not have much technological and software experience. An app that allows her to assist while she is working by organizing support or donating.

### ► FRUSTRATIONS

Confusion with current hurricane relief systems. Sometimes becomes too busy with work to meaningfully contribute. Struggles with fundraising and finding support for victims after the initial hurricane is over.

## Victim

# ETHAN CARTER

*"I want to know the best ways to take care of my family and pets during a emergency."*



### QUICK FACTS

-  Job: Retail Employee
-  Age: 83 years old
-  Gender: Male
-  Address: 1842 Pinewood Drive, Columbia, SC 29205

### ABOUT ME

#### Lifestyle

Majority of time spent working, dealing with call outs and missing employees. Lives a simple, budget-conscious lifestyle, works full-time in retail.

#### Personality

Friendly, hardworking, easygoing, and independent. He can get stressed about finances but generally stays optimistic and practical.

#### Hobbies

Hanging out with Friends, Fishing, Spending Time at the Beach, Playing Video Games

### EDUCATION & EXPERIENCE

#### ✓ Education

High School Diploma

#### ✓ Technology Proficiency

still uses a landline, desktop in household, has a cell phone with limited functional knowledge

#### ✓ Technology Platforms

Google, SMS Texting, Calling

### GOALS & FRUSTRATIONS

#### ► GOALS AND NEEDS

##### Goals:

Managing Financial Resources after Emergency  
Have Reliable Exit Strategy  
Shelter/Food/Water Availability

##### Needs:

Reliable Transportation from Site for Elderly  
Medical Assistance  
Financial Resources for Damage  
Food/Shelter/Water Availability

#### ► FRUSTRATIONS

Car Issues/Unreliable Transportation  
Limited Savings  
Stressed Financial Situation  
Minimum Reliable Social Connections  
Repairing Damage After Emergency  
Managing Pets\Dependents After\During Accident

## First Responder

# JAMIE WOODS

*"I need immediate and organized access to important information when responding to emergencies."*



### QUICK FACTS



Job: EMS Paramedic



Age: 27



Gender: Female



Address: 100 Castle Rd,  
Columbia, SC 29210

### ABOUT ME

#### Lifestyle

Works long and unpredictable shifts, used to high pressure situations and making quick decisions

#### Personality

Empathetic, organized, practical, works well under pressure, decisive, team-oriented

#### Hobbies

Running, watching documentaries and reality TV, traveling with friends and family, photography

### EDUCATION & EXPERIENCE

#### ✓ Education

Associate degree in Emergency Medical Services

#### ✓ Technology Proficiency

High- regularly uses dispatch software, GPS, tablets, and mobile apps at work

#### ✓ Technology Platforms

Work-issued computer, personal ipad, iphone

### GOALS & FRUSTRATIONS

#### ► GOALS AND NEEDS

##### Goals:

Quickly see emergency requests, their urgency, and where the victim is located. Coordinate with other responders. Know which shelters/hospitals have capacity and how to get there.

##### Needs:

Real-time emergency requests organized by priority  
Victim location with GPS navigation and road closures  
Shelter and hospital status  
Ability to update request status

#### ► FRUSTRATIONS

Incomplete information from callers  
Outdated road closure information  
Poor cell service  
Duplicate emergency reports  
Difficulty organizing and coordinating medical aid during natural disasters

## 3. Constraints

### Time Constraints

- The system must be designed and developed within the timeframe established for the course project.

### Monetary Constraints

- The project's development has been allocated a budget of \$0.
- The system should rely on free resources.

### Technical Constraints

- The code must be done in Java.
- The website must be accessible on a variety of different devices including PC, laptops, and mobile devices.

## 4. Overall Description

The Hurricane Emergency Response and Relief System will provide a centralized resource for hurricane response and recovery. It will support affected residents, emergency responders, shelter personnel, volunteers, and relief organizations by connecting users with emergency assistance and longer-term recovery resources. The system will work alongside existing emergency and relief services and may connect users with information from outside organizations. It will be coded using java and accessible through devices like smartphones and computers.

## 5. Business Use Cases



## 6. Functional Requirements

[Requirements Spreadsheet Link](#)

## 7. Non-Functional Requirements

### Usability

- The system shall allow a user to begin a relief request before entering optional request information.
- The user interface shall support interaction on a mobile-sized display.
- The user interface shall support interaction on a desktop-sized display.
- The system interface shall support accessible interaction for users with visual, hearing, cognitive, or mobility limitations.

### Connectivity and Availability

- The system shall provide access to previously stored emergency information when an internet connection is unavailable.
- The system shall provide previously stored shelter information when an internet connection is unavailable.
- The system shall provide previously stored hurricane-preparation information when an internet connection is unavailable.

### Security & Verification

- Volunteers who respond at private residences shall complete the required verification process before being assigned to requests.
- Professional credentials recorded for specialized volunteers shall be validated before those credentials are used for request matching.
- Registered accounts shall require authentication before users access account-specific information.

### Language

- The system shall provide an English-language interface.
- The system shall provide a Spanish-language interface.

## 8. Definitions and Acronyms

**EMS:** Emergency Medical Services; medical responders who provide urgent care and transport during an emergency.

**First Responder:** A trained emergency worker, such as an EMS worker, firefighter, or police officer, who responds to urgent situations.

**Hurricane Victim:** A person affected by a hurricane who may need emergency assistance, shelter, supplies, or recovery support.

**Emergency Shelter:** A temporary safe location that provides housing and basic necessities to displaced residents.

**Relief Resource:** Supplies or services available to assist hurricane victims, such as food, water, medical care, transportation, or cleanup assistance.

**Recovery Resources:** Services that help individuals and communities recover after the immediate emergency, including housing, rebuilding, emotional support, and debris removal.

**Volunteer:** A person who offers time and assistance to support relief or recovery efforts.

**Shelter Administrator:** An authorized user responsible for managing shelter information, capacity, supplies, and needs in the system.

**System Administrator:** An authorized user responsible for managing user access, maintaining system information, and overseeing the system.

**FEMA:** Federal Emergency Management Agency; a United States government agency that coordinates federal disaster response and recovery assistance.

**Red Cross:** The American Red Cross, a humanitarian organization that provides disaster relief, emergency assistance, and preparedness resources.

**Hazard:** A dangerous condition that may affect safety or travel, such as flooding, a blocked road, a downed tree, or a downed power line.

## 9. Competitive Analysis

|                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>Tidal Basin Group</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Strengths</b>                                                                  | <p>The website organizes important sections such as preparing, responding, recovering, and reducing future disaster damage which allows for easy access and a clean interface. It also provides a Hurricane Resource center with recovery information and how people should prepare for hurricanes.</p> <p>The website presents a professional design with descriptions of its experience, service areas, team members, and past work. This can boost confidence in the government agencies and relief organizations in their decision to contact and work with the Tidal Basin Group.</p> <p>It features a big red button titled “Reach Out” making it super clear how people can ask questions, and give feedback to the organization.</p> |

|                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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|  | <h2>Tidal Basin Group</h2>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <h3>Weaknesses</h3>                                                               | <p>The website is mostly designed for government agencies, organizations, and potential business clients opposed to individual hurricane victims. For example, if there was a resident needing immediate shelter, food, transportation, or medical help it may be difficult to get direct assistance through the site.</p> <p>Individual residents cannot communicate with responders through the website as well as submit for their own emergency requests.</p> <p>The website is also lacking a live, local map for residents for them to quickly and easily see shelter availability, road closures, supply locations, or nearby emergency services.</p> |
| <h3>Audience/Focus</h3>                                                           | <p>The Tidal Basin Group website is primarily focused on government agencies, emergency management organizations, and communities that need professional disaster planning, response, and recovery support. It does not focus much on the individual victim of a hurricane making it hard for users to get help for themselves or their families. The main focus is to connect the organizations with disaster-management services, including FEMA assistance, housing recovery, grant management, and debris management.</p>                                                                                                                                |

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|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>American Red Cross</b> | <h2>American Red Cross</h2>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <h3>Strengths</h3>                                                                                          | <p>As an organization, the American Red Cross has an immense amount of medical, nutritional, and emergency resources available in every part of the United States. These resources are readily available via the Red Cross's extensive logistical system including 230 local chapters that are emergency ready at all times.</p> <p>Training and education are a powerful strength of the American Red Cross. Across the country, the Red Cross offers basic medical, survival, and emergency response training for volunteers and citizens.</p> <p>Finally, the volume of services the organization offers support each other, as well as operate individually. For instance, Blood and Platelet Donations support Disaster Relief and Response in medical emergency situations. This organizational structure allows the American Red Cross to reduce their overhead costs allowing them to grow larger than other nonprofit organizations.</p> |

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|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Weaknesses</b>     | <p>Most of the weaknesses that the American Red Cross face are due to the size of the organization. The scale of the organization causes a lot of bureaucracy when responding to disasters. Relying on volunteers creates inconsistencies in training, mobility issues, and logistical problems. Finally, since the Red Cross responds to emergencies across the country, the organization can face multiple national level emergencies at one time which will heavily drain the organizations resources.</p> <p>The American Red Cross website is dependent on the internet, therefore a victim couldn't request help or reach information in an emergency if they do not have internet access.</p> |
| <b>Audience/Focus</b> | <p>Legislators, Medical Professionals, Emergency Response Professionals, First Responders, Physically Healthy Volunteers, Citizens with Medical Training, Volunteers for Blood and Platelet Donations, Potential Victims.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

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|  <b>FEMA</b> | <b>FEMA</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Strengths</b>                                                                              | <p>Before a disaster, FEMA provides citizens with disaster readiness information and training, flood insurance, and flood risk information. They work with state and local governments to help them plan and prepare for emergencies. They also provide grants to research effective disaster prevention and mitigation.</p> <p>Being a government agency, FEMA can coordinate natural disaster response with states, tribes, non-profits, private organizations, and other federal agencies. During a disaster, a state government can put in a request to FEMA for support. Once approved by the president of the United States, FEMA helps with providing supplies, temporary housing, infrastructure repair, and search and rescue teams.</p> <p>In addition to helping with immediate disaster response, FEMA also supports communities' long-term recovery. They help communities rebuild homes and public infrastructure, and prevent further disasters. Individuals can request financial assistance on <a href="https://www.disasterassistance.gov">DisasterAssistance.gov</a>.</p> |

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Weaknesses</b></p>     | <p>FEMA's main weaknesses are a result of the complexity of the federal government. It has to coordinate with multiple levels of government and follow processes and regulations that other types of organizations don't have to. This can slow down response times.</p> <p>Additionally, FEMA is not designed to replace, but rather support, state and local emergency response services. Communities still need to have strong response and relief systems of their own.</p> <p>If FEMA has to attend to multiple disasters at the same time, their resources can be stretched thin. Due to climate change, weather-related natural disasters are only becoming more common.</p> <p>Finally, like the American Red Cross, FEMA accepts support requests for individuals through its website. If a natural disaster blocks a victim's access to the internet, they can't request help.</p> |
| <p><b>Audience/Focus</b></p> | <ul style="list-style-type: none"> <li>● Citizens in at-risk areas</li> <li>● Natural disaster victims</li> <li>● State, territory, tribal, and local governments</li> <li>● Potential employees</li> <li>● Providers of aid supplies</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## Summary

|                                                      | Strengths                                                                                                                                                                                        | Weaknesses                                                                                                                                                                             | Audience/Focus                                                                                                                                                                                  |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Competitor 1</b><br><br><b>Tidal Basin Group</b>  | <ul style="list-style-type: none"> <li>- Easy and clear interface</li> <li>- Plenty of resources</li> <li>- Professional credentials</li> </ul>                                                  | <ul style="list-style-type: none"> <li>- Not individualized</li> <li>- Lacking map, road closures, and more for during the hurricane.</li> </ul>                                       | <ul style="list-style-type: none"> <li>- Focuses more on big companies and organizations</li> <li>- Focuses a lot on after effects of the hurricane and less materials for during it</li> </ul> |
| <b>Competitor 2</b><br><br><b>American Red Cross</b> | <ul style="list-style-type: none"> <li>- Vast Amount of Resources</li> <li>- Chapters Across The County</li> </ul>                                                                               | <ul style="list-style-type: none"> <li>- Too large, creates bureaucracy</li> <li>- Manages Emergencies nationally leaving resources spread thin</li> </ul>                             | <ul style="list-style-type: none"> <li>- Focuses on getting volunteers to train and respond to emergencies</li> <li>- Provides victims with emergency resources</li> </ul>                      |
| <b>Competitor 3</b><br><br><b>FEMA</b>               | <ul style="list-style-type: none"> <li>- Powerful organization with government presence</li> <li>- Long term recovery</li> <li>- Maintains presence in communities across the country</li> </ul> | <ul style="list-style-type: none"> <li>- Complex organizational structure causing bureaucracy</li> <li>- Does not operate a replacement for emergency services just support</li> </ul> | <ul style="list-style-type: none"> <li>- Citizens</li> <li>- Victims</li> <li>- Smaller Governments</li> <li>- Potential Employees</li> <li>- Aid Suppliers</li> </ul>                          |

## Overall

Within the hurricane response and relief system there are many organizations that have direct, supplemental, and indirect roles in rescuing victims, organizing resources, and repairing communities. There are three distinct types of organizations involved in the relief system including governments, nonprofits, and private corporations.

The organizations that were analyzed for the purpose of this document focused on the rescuing and retrieval of individuals. This includes shelter, nutritional, and medical logistics, as well as the deployment of emergency services. However, through our analysis our team identified an overall weakness within the current system. There is very little focus on long-term recovery for individuals, communities, and smaller businesses within disaster zones. While there is some focus on recovery, mostly in the private sector, the need is not satisfied. There are many cases where these groups are left with damaged structures, limited resources, and few options for reconstruction.

Therefore, the development of a new hurricane relief system would be most effective if it focused on supporting individuals, communities, and small businesses with a long form capacity with intention to repair the area to its predisaster state.

## 10. References

Tidal Basin Group. *Hurricane Resource Center*.

<https://www.tidalbasingroup.com/education/resource-hubs/hurricane-resource-center/>

American Red Cross. *Hurricane Preparedness*.

<https://www.redcross.org/get-help/how-to-prepare-for-emergencies/types-of-emergencies/hurricane.html>

Federal Emergency Management Agency. *Hurricane Preparedness and Evacuation*

*Planning*. <https://www.fema.gov/emergency-managers/risk-management/hurricanes>

## 11. Appendices

### **Appendix A - Business Use Case Diagram**

This appendix contains the business use case diagram for the Hurricane Emergency Response and Relief System. It identifies the primary actors—Victim, Volunteer, EMS Worker, Shelter Administrator, and System Administrator—and their high-level interactions with the system.

### **Appendix B - User Personas**

This appendix contains the user personas developed for the Shelter Volunteer, Hurricane Victim, and First Responder. The personas help explain the needs and goals of representative users and what someone might look like who will utilize our app or website.